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RESEARCH ARTICLE

Fed-DPSDG-WGAN: Differentially Private Synthetic Data Generation for Loan Default Prediction via Federated Wasserstein GAN

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ABSTRACT Lenders typically conduct thorough credit checks to mitigate credit default risk before lending money. Proactively predicting loan defaulters has become increasingly important. However, creating robust deep-learning algorithms that classify loan defaulters requires abundant data, potentially compromising individuals' privacy. We investigate the potential of replacing the original data with statistically similar synthetic data to be used as training data to predict loan defaulters. Generative adversarial networks stand out with the best privacy utility trade-off among the many approaches that can generate synthetic data. However, limited data and membership inference attacks still significantly affect the privacy of the synthetic data generated. This paper uses federated learning to create synthetic data to improve loan defaulter prediction. By allowing multiple institutions to collaboratively train a shared model without directly exchanging sensitive loan data, this privacy-preserving approach allows for the utilization of more extensive and diverse datasets, potentially leading to significant improvements in prediction accuracy compared to traditional, siloed approaches. This method also builds a layer of privacy to safeguard against membership inference attacks using a differentially private optimizer layer in the WGAN architecture. The proposed framework utilizes a Min-Max Module and Autoencoders for global representation of distributed loan data, followed by a Federated WGAN, enhanced with a Gaussian noise-added Adagrad optimizer, to generate high-fidelity, privacy-preserved synthetic credit risk data. Furthermore, we use CTGAN to mitigate class imbalance in the synthetic dataset. Finally, we test the generated synthetic dataset using a model trained on the real dataset to check its utility. With this novel framework, we could attain a privacy-utility tradeoff in a federated environment. Our extensive experimentation, which includes a variety of communication rounds, client populations, and privacy budgets, demonstrates that the proposed framework achieves an impressive 95% accuracy. The proposed model marks a significant improvement over both a non-federated setup, which achieves 82% accuracy, and FLIGAN with 73% accuracy, which possesses the same functionalities with different techniques.

INDEX TERMS Federated learning, WGAN, CTGAN, differentially private data, oversampling, trivial auto-encoder, Gaussian noise.

I. INTRODUCTION

Credit risk, the likelihood of a borrower failing to repay their loan, presents a significant challenge for financial institutions. Accurate prediction of such risk during the loan approval process, also known as Loan Defaulter Prediction, is crucial to minimize potential losses and maintain profitability [1]. Researchers have explored various Machine Learning and Deep Learning techniques to predict the loan defaulters in advance and address this problem [2], [3], [4], [5]. However, training models with original data creates significant privacy and legal risks. Consequently, developing privacy-preserving techniques is essential for responsible and ethical loan defaulter prediction.

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Deep generative models offer a promising solution for anonymizing the dataset by creating synthetic datasets that can be used for various purposes without revealing sensitive information about individuals in the original database. These models, including GANs [6], VAEs [7], and Diffusion Models [8], have demonstrated effectiveness in synthesizing data in various applications, from image generation [9] to tabular data [10]. Numerous researchers have used GANs to generate statistically similar synthetic data to address data privacy challenges. [11], [12].

However, GANs suffer from the threat of attacker models like model inversion attacks and membership inference attacks. Researchers have also explored the combination of GANs with differential privacy to address privacy concerns [13], [14], [15], [16]. Deep generative models also succeed more in recreating continuous rather than discrete data. We have explored a unified preprocessing layer to address this challenge.

Furthermore, utilizing vast and heterogeneous data sets from various financial institutions improves credit risk evaluation and predictive accuracy. However, sharing sensitive customer data among financial institutions to enhance predictive models presents significant privacy risks and challenges.

The fact that financial data often exhibits non-IID (non-independent and identically distributed) characteristics further complicates this challenge [17]. In IID data, all random variables are independent and have the same probability distribution. In contrast, non-IID data feature non-independent variables with varying label distributions across different data providers. This heterogeneity poses significant challenges for machine-learning approaches to model and generate synthetic data. Combining Generative Adversarial Networks (GANs) with Federated Learning (FL) addresses the challenges of data inconsistency and the GDPR of not sharing data. Federated Learning enables financial institutions to train machine learning models collaboratively, masking the individual's data [18].

This combination of Generative Adversarial Networks and Federated Learning is a promising approach for generating differentially private synthetic data in a distributed environment. Introducing differential privacy in the optimizer layer further protects Generative Networks from model inversion attacks and membership inference attacks.

Addressing the challenges of data representation in GANs, differentially private synthetic data generation, handling non-IID ness of data, mitigating model inversion and membership inference attacks, we introduce the Differentially Private Synthetic Data Generation for Loan Defaulter Prediction via Federated Wasserstein GANs (Fed-DPSDG-WGAN) framework. This framework integrates a Unified Data Representation using Autoencoder, federated WGAN with Gaussian noise-added Adagrad optimization, and oversampling techniques to enhance model performance without compromising sensitive information.

We summarize the contribution of our work as

1) **Unified Data Representation:** We introduce a novel Unified Data representation framework to standardize mixed data types and preprocess all the client's data.
2) **WGAN with Differentially Private Optimizer Layer:** A WGAN framework incorporating a Gaussian noise-enhanced Adagrad optimizer to generate high-fidelity synthetic loan data while preserving privacy. It safeguards the model from any inversion or inference attacks
3) **Federated WGAN + Adagrad Optimizer:** A WGAN with Gaussian noise added Adagrad optimizer framework is integrated into a Federated Environment. This approach addresses research challenges such as handling non-IID data, enhancing model performance, and ensuring data privacy.
4) **Oversampling:** An oversampling technique is integrated into the framework to balance the generated imbalanced synthetic dataset, thereby improving model accuracy.
5) **Experimental Results:** Finally, this paper presents extensive experimental results to assess the performance of our proposed framework compared to other state-of-the-art models. Our framework proved superior to all other state-of-the-art methods in a comprehensive evaluation.

II. RELATED WORK

Federated learning is a decentralized approach to machine learning that allows multiple parties to collaboratively train models without sharing their raw data. This technique primarily benefits data privacy by keeping the data on local devices and exchanging only model updates.

Later, researchers adapted federated learning principles to generative models, specifically Generative Adversarial Networks. This approach enables the training of GANs on distributed, non-independent, and non-identically distributed (non-IID) data sources, effectively addressing challenges related to communication and privacy. While Federated GAN research has primarily concentrated on image data, its applicability extends to other data types, such as the tabular and time-series data used in our experiments.

This section explores the research contributions of Federated GANs, which form a solid foundation for the current study. Table 1 provides a concise overview of existing research on Federated Generative Adversarial Networks (Federated GANs) specifically applied to image datasets. Later researchers have extended the application of Federated Learning with GANs on tabular datasets, such as Credit Card, Adult, Loan, Intrusion, and Cover Type, and Table 2 explicitly summarizes such papers.

Fed-TGAN represents the inaugural framework for federated learning designed explicitly for Tabular Generative Adversarial Networks. It dealt with non-IID data among clients. It surpasses two other decentralized architectures,

TABLE 1. Contributions of federated GANs on image dataset.

Paper	Model	Dataset	non-IID
FedCSCD-GAN [19] Rehman et.al 2024	Cramer GAN	Prostate and Breast Cancer	No
UFed-GAN [20] Wijesinghe et al.2023	DCGAN	FMNIST,SVHN,CIFAR10	Yes
IFL-GAN [21] Li et al.2023	DCGAN	CIFAR10, MNIST, SVHN	Yes
FedMed-GAN [22] Wang et al.2023	GAN	IXI, BRATS2021	Yes
PP-FedGAN [23] Ghavami et al.2023	WGAN	MNIST,SVHN	No
FL-GAN + HE [24] Pejic et al.2022	GAN	CIFAR10	No
EFFGAN [25] Ekblom et al.2022	DCGAN	MNIST,FMNIST	Yes
FedSys [26] Behera et al.2022	GAN	MNIST,FMNIST,CIFAR10	Yes
FedDPGAN [27] Zhang et al. 2021	DP-GAN	COVID-19	Yes
FCGAI [28] Zhou et al.2021	GAN	Air Pollutants	Yes
FedGAN [29] Rasouli et al. 2020	CGAN	MNIST,CIFAR10,CelebA	Yes
FeGAN [30] Guerraoui et al. 2020	CGAN	MNIST,FMNIST	Yes

TABLE 2. Contributions of federated GANs on tabular dataset.

Paper	Model	Dataset	non-IID
FLIGAN [31] Mali et al.2024	GAN	Albert,Bank Adult,Creditcard,Intrusion	Yes
GTV [32] Zhao et al.2023b	CT-GAN	Adult,Covtype, Loan Intrusion, Credit	Yes
GDTS [33] Zhao et al.2023a	GAN	Adult, Covertype,Credit, Intrusion	Yes
HT-Fed-GAN [34] Duan et al.2022	CCGAN	Adult,Health,CovType, Intrusion,Body	Yes
Fed-TDA [35] Duan et al. 2022	Data Augmentation	CovType,Clinical,Tuber Intrusion,Body	Yes
Fed-TGAN [36] Zhao et al.2021	GAN	Adult, Covertype,Credit, Intrusion	Yes
GS_WGAN [[37]] chen et.al.2020	WGAN	MNIST and Fashion	Yes

MD-TGAN [38] and vanilla FL-TGAN, as well as a centralized method. Fed-TGAN has two key features: (i) privacy-preserving feature encoding, which facilitates model initialization across heterogeneous data sources, and (ii) table-similarity-aware weighting, used for merging local models. However, this model fails to address the bias of class imbalance in the model training.

In the paper HT-Fed-GAN, researchers highlighted their method as a pioneering solution for privacy-preserving data synthesis in horizontally partitioned decentralized tabular datasets. It uses Fed-VB-GMM to handle mode collapse and incorporates differential privacy to protect against private data leakage during federated training.But the model does not handle imbalanced data. GDTS Employs an individual GAN for each client, enabling personalized model training that captures distinct local data characteristics. This approach outperforms adaptations such as MD-GAN by effectively managing diverse data distributions across clients. However, these researchers failed to acknowledge the need for a universal preprocessing technique to standardize the data and make learning easy for the GAN models. These frameworks are still prone to membership inference attack in their central server where the model is stored. In the meantime, Duan et al. 2022 introduced Fed-TDA [35], a federated tabular data augmentation that tackles the non-IID problem in federated learning. Unlike GAN-based methods, Fed-TDA synthesizes tabular data using simple statistics, like column distributions and global covariance. The technique employs multimodal distribution transformation (MDT) and inverse cumulative distribution mapping (ICDM) for discrete columns. Autoencoder-based data representation can offer more robust modeling of discrete values compared to MDT. Autoencoders achieve this by learning complex relationships within the data, demonstrating resilience to noise [39].

Though Fed-TGAN and HT-Fed-GAN address the challenge of non-IID tabular data in a federated environment, they

still encounter issues such as the computational complexity of Gaussian mixture models (GMMs) and intricate parameter tuning. They do not deal with incomplete data.

Researchers in the paper titled FLIGAN offer an efficient solution for synthetic data generation in federated settings, particularly for tabular data with incomplete information. It uses a Federated Wasserstein GAN (WGAN) with class-wise sampling and node grouping to reduce computational demands, simplify the process, and preserve privacy. Studies demonstrate that this approach outperforms other tabular-based GANs. This method emphasizes privacy preservation, synthetic data generation, and handling incomplete data, outperforming all other tabular-based GANs.

As we look deeper into the history of research on predicting financial frauds, it is clear that many researchers have worked on credit card fraudulent prediction applications, and not many on loan default prediction. Researchers have not developed an innovative method for generating differentially private synthetic data to identify loan defaulters in a decentralized environment. To address this, we need to introduce a framework that can generate differentially private synthetic data in a distributed environment for predicting loan defaulters, focusing on maximizing the privacy-utility trade-off.

III. MATERIALS AND METHODS

A. MATERIALS

1) DATASET

The Home credit loan risk data set [40] is a comprehensive resource designed to evaluate the borrower's default probability. It draws information from several key datasets, including the Bureau and Bureau Balance datasets containing information on previous loans and outstanding balances from other institutions, the earlier applications and installment payments datasets detailing prior home loans and payment history, and the Cash Balance and Credit Card Balance datasets describing remaining and outstanding credit card balances. This diverse data collection makes the dataset highly data-dependent and non identically distributed (non-IID). The data architecture diagram in Figure 1 depicts the relational data model of the datasets. The dataset comprises 3,07,511 records across 122 columns, encompassing both categorical and numerical features. The primary target variable is binary, signifying whether a borrower has Credit Defaulters (1) or Not Credit Defaulters (0). The dataset is entirely unbalanced, with 91.9% of records having values of 0 or No credit risk customers and 8.07% having values of 1 or credit defaulters. Table 3 describes the Home credit loan risk data set.

2) NON-IID DATA: DEFINITION

A key challenge for federated learning is handling non-independent and identically distributed (non-IID) data [41], a common issue in finance where institutions often have diverse datasets. While FL performs best with IID data, real-world data is often non-IID. This heterogeneity can cause model divergence, particularly with parametric models in horizontal FL. A frequent type of non-IID data is label distribution skew, where label distributions vary across clients. This paper focuses on simulating this skew, often resulting from localized data collection where clients gather similar, yet label-skewed training data [42].

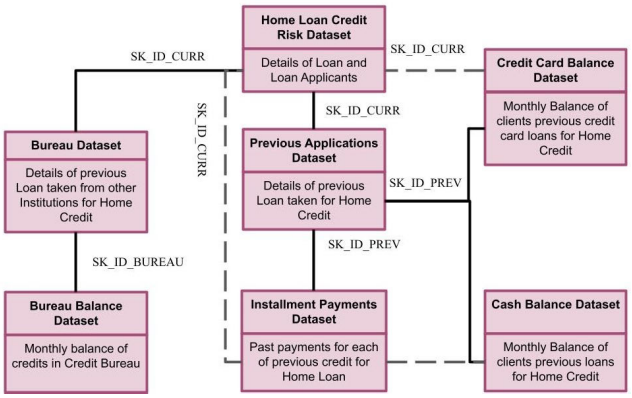


FIGURE 1. A comprehensive overview of the home loan credit risk dataset.

TABLE 3. The dataset description.

Total number of Records	3,07,511
Number of Categorical Attributes	16
Number of Numerical Attributes	106
Number of Target Attributes	1
Target Attributes	Credit Defaulters(1) /Not Credit Defaulters(0)

B. METHODS

This section explores the background of the techniques and concerns addressed in this paper.

1) ATTACK MODELS

Unlike centralized machine learning, Federated learning trains models on multiple clients, keeping data private and localized. However, FL is vulnerable to attacks, especially membership inversion attacks and model inversion attacks, which are a serious threat.

1) **Membership Inference Attacks:** Membership Inference Attacks (MIAs) pose a significant privacy risk, particularly in sensitive domains like finance, even when using privacy-aware techniques like Federated Learning [43]. Membership Inference Attacks (MIAs) actively reveal whether a specific individual's data was used to train a machine learning model, potentially exposing confidential information.
Functional test are the most common type of Membership Inference attack. A functional MIA attack occurs when an attacker can only interact with a machine-learning model by providing inputs and observing the outputs. Lacking knowledge of the

model's internal structure, the attacker must infer sensitive training data by strategically querying the model [44]. We simulated this functional attack to test the MIAs with varying privacy budget.

2) **Model Inversion Attacks:** Model Inversion Attacks are a significant security concern for federated learning. These attacks let adversaries reverse-engineer a model to infer private information about the training data. Among the various methods suggested for risk mitigation, Differential Privacy (DP) stands out as a key solution that provides a robust framework for protecting individual privacy by adding controlled noise at various levels.

This paper introduces controlled noise at the optimizer level during the proposed WGAN model training to defend against membership inference and model inversion attacks. [44].

2) DIFFERENTIAL PRIVACY USING GAUSSIAN NOISE

Cynthia Dwork and colleagues introduced Differential Privacy (DP) technology in 2006 in a paper titled "Differential Privacy" [45]. The core idea behind differential privacy is to add noise during data release, making it difficult for attackers to accurately infer an individual's sensitive information. Dwork et al.'s work on differential privacy revolves around the Equation 1. Recently, Local Differential Privacy (LDP) [46] has become widely accepted as the standard for ensuring privacy when collecting user data from Edge devices, but LDP's approach of individual data perturbation, designed to maximize privacy on device data collection, inherently generates more noise, leading to lower data utility relative to DP, where post-aggregation noise addition preserves greater accuracy. Therefore, our proposed model must strike a balance between privacy and utility, making Differential Privacy (DP) a more appropriate perturbation technique than Local Differential Privacy (LDP).

$$M : D \longrightarrow R, \tag{1}$$

where D represents the data retrieval domain, and R is the range of the dependent variable. To ensure the mechanism satisfies ϵ-differential privacy, it must hold for any two adjacent inputs b,b' ϵ D, and every subset of outputs $S \subset R$, following the prescribed mathematical condition.

Between the Laplacian and the Gaussian type of noise addition, Gaussian noise-added differentially private data seems more relevant to a deep learning scenario. If S_f is the sensitivity of the random function for the two adjacent datasets b, b', f(b) is the query function, and ${S_f}^2.\sigma^2$ is the variance of the Gaussian distribution. For each pair of adjacent inputs b and b', the Gaussian noise mechanism can be expressed in Equation 2.

$$M(b) = f(b) + N(0, {S_f}^2.\sigma^2) \tag{2}$$

where $N(0,{S_f}^2.\sigma^2)$ is the noise to disturb the distribution in 0 and the standard deviation $S_f.\sigma$. Then, we define the sensitivity S_f of the random function f in Equation 3.

$$\Delta f = max_{b,b0}||f(b) - f(b')||_2 \tag{3}$$

For each pair of adjacent inputs, b and b', the Gaussian noise mechanism can be expressed in Equation 3. The Gaussian mechanism satisfies (ε,δ)-differential privacy. It asserts that the function F(x), as defined below, satisfies Equations. 4 and 5.

$$F(x) = f(x) + N(\sigma^2) \tag{4}$$

where

$$\sigma^2 = \frac{2s^2 \log(1.25/\delta)}{\epsilon^2} \tag{5}$$

Here, s represents the sensitivity of the function F. $N(\sigma^2)$ denotes sampling from a Gaussian distribution centered at 0 with variance σ^2 [12].

Adding Gaussian noise at the optimizer level hinders attackers' ability to infer training set membership or reconstruct specific data points, thus mitigating model inversion and membership inference attacks.

3) TRIVIAL AUTOENCODER

A Trivial Autoencoder (TAE) is a neural network designed to learn a compressed representation of input data and reconstruct the original input as accurately as possible [47]. It consists of an encoder h=f(x) and a decoder that reconstructs r=g(h). To ensure that the reconstructed data closely resembles the original input, the training process should minimize the loss function given in Equation 6.

$$L(x, g(f(x))) \tag{6}$$

where L penalizes the dissimilarity between the reconstructed data, g(f(x)), and the original input, x. Unlike traditional autoencoders, a key characteristic of TAEs is that the middle layer's size, h, equals the input data size, x. This layer's hyperbolic tangent (tanh) activation function effectively transforms binary input data into continuous values.

4) FEDERATED LEARNING

Federated learning enables multiple entities to train a shared model collaboratively without exchanging raw data, as shown in Figure 2. This collaborative approach is particularly beneficial for privacy-sensitive applications with de-centralized data.

Formally, let $\mathcal{C} = \{C_1, C_2, \ldots, C_K\}$ represent the set of participating entities, each holding a local dataset D_i. The objective is to train a global model θ by aggregating local updates while preserving data privacy. This concept can be formalized mathematically using Equation 7

$$\min_{\theta} F(\theta) = \frac{1}{K} \sum_{i=1}^{K} f_i(\theta) \tag{7}$$

For collaborative learning, participants compute and share local gradients, enabling the aggregation of global model

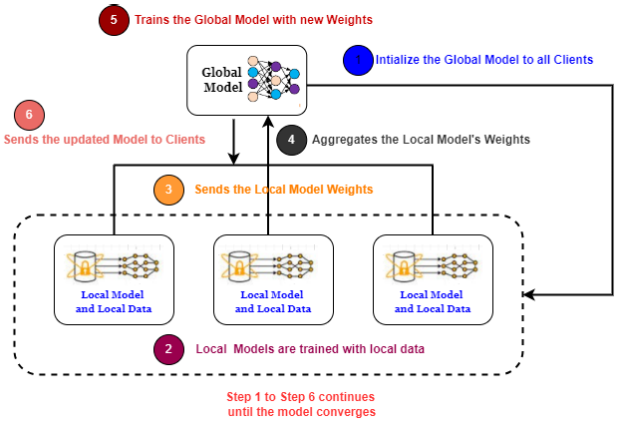


FIGURE 2. An overview of federated learning process.

parameters. The iterative update at round r is given by Equation 8

$$\theta_{r+1} = \theta_r - \eta \sum_{i=1}^{K} \nabla f_i(\theta_r) \tag{8}$$

Here, η denotes the learning rate [48]

5) FEDAVG

One of the earliest and most widely used methods in FL is FedAVG (Federated Averaging) [49]. During the aggregation, the parameters of each client are weighted and averaged to produce a global model. The model is trained iteratively. let θ_t^g denote the latest global model aggregated by the central server at iteration step t, and m_i (for $i = 1, 2, 3, 4, \ldots, K$) denote the devices of all K clients. The central server first broadcasts θ_t^g to all m_i, and then every device (say the k-th) initializes θ_t^i as θ_t^g. Then, the i-th client performs E (where $E > 0$) local updates as illustrated in Equation 9

$$\theta_t^i(k+1) = \theta_t^i(k) - \eta_k \nabla \ell \left(\theta_t^i(k), x_i, j, y_i, j\right) \tag{9}$$

where $k = 1, 2, 3, \ldots, E$ denotes the local training step, and ℓ is the classification loss function used by the i-th client, while $x_{i,j}, y_{i,j}$ (where $j = 1, 2, 3, \ldots, |V_i|$) are the training instances hosted locally by the i-th client. Finally, $\theta_t^i(k)$ is the local model updated at the k-th step of stochastic gradient descent performed locally; note that $\theta_t^i(k) = \theta_t^i = \theta_t^g$ when $k = 0$. After the clients finish training their local models, they send their parameters to the central server. The global model is derived at the central server and is then aggregated by averaging the local models as in Equation 10

$$\theta_{t+1}^g = \sum_{i=1}^{K} \theta_t^i(E) \tag{10}$$

Note that $\theta_t^i(E)$ denotes the models trained locally after E (where $E > 0$) rounds of gradient descent [50].

6) TABULAR DATA USING CONDITIONAL GAN (CTGAN)

Conditional Tabular GAN (CTGAN) is among the best techniques for modeling imbalanced tabular data. We employed this technique to perform oversampling of the minority class. It generates samples that enhance the performance of machine learning models for binary classification. GAN-powered augmentation can alter the importance scores of certain features, influencing the learning dynamics of classifiers. As a result, classifiers trained with augmented data can give more balanced attention to each class, leading to a more uniform distribution in the confusion matrix during inference [51]. This is an essential step in our framework because the dataset is highly skewed.

IV. PROPOSED FRAMEWORK

In the rapidly evolving financial services landscape, robust data-driven decision-making is paramount. However, this reliance on extensive customer data raises significant privacy concerns and regulatory challenges, particularly in light of stringent data protection laws such as the General Data Protection Regulation (GDPR). This paper presents a novel framework designed for generating privacy-preserved synthetic data for loan defaulter prediction, employing advanced deep learning techniques in both centralized (Figure 3) and collaborative (Figure 4) Environments.

The proposed framework utilizes a three-stage process: a unified approach for effective data representation, WGANs (enhanced with Gaussian noise added Adagrad optimization) for generating differentially private synthetic data that prevents both model inversion and membership inference attacks, and Conditional GANs (CTGAN) for dataset balancing.

In a centralized environment, the framework operates as follows: the system initially uses the entire dataset to train the autoencoder, generating a continuous data representation. The framework feeds this representation into the WGAN with Gaussian noise-added adagrad optimizer, which produces differentially private synthetic data. Finally, CTGAN can be employed to balance the resulting synthetic dataset. The architecture of the proposed framework in a centralized environment is demonstrated in Figure 3.

Training the WGAN model with limited data poses significant challenges. The model struggles to capture complex relationships, leading to overfitting. This results in poor generalization and reduced predictive accuracy on unseen data." Hence, moving to a Federated approach is the best way to tackle this issue.

Unlike centralized approaches, a collaborative environment utilizes Federated Learning (FL) to allow multiple banks to engage in joint modeling efforts without centralizing sensitive data. FL inherently safeguards data privacy by training machine learning models directly on local devices, ensuring that data remains on users' devices and minimizing the risk of breaches and unauthorized access. This decentralized approach promotes shared learning while

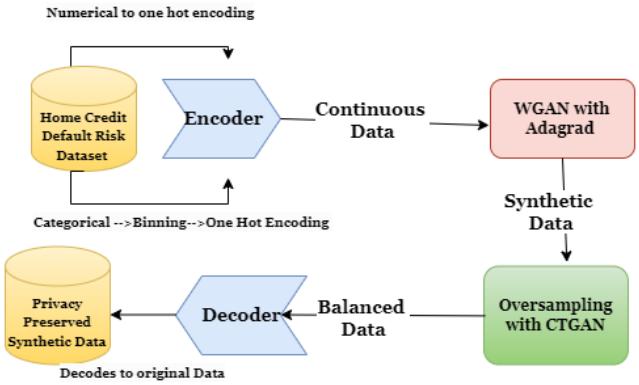


FIGURE 3. The differentially private synthetic data generation architecture in centralized environment.

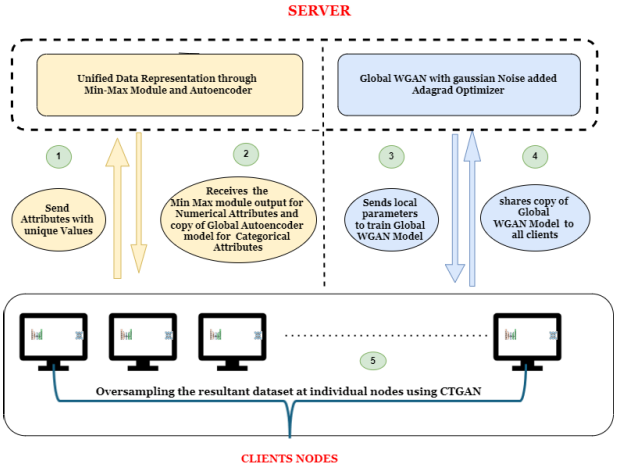


FIGURE 4. The proposed differentially private synthetic data generation architecture in federated environment.

preserving data privacy across institutions and the proposed decentralized approach is depicted in Figure 4.

The proposed research aims to tackle all the challenges in a de-centralized environment, including unified data representation, handling heterogeneous and imbalanced data, and two specific attacks: model inversion and membership inference attacks.

The methodology works as follows

A. INITIALIZATION OF THE CLIENTS

The first step in the framework is defining the collaborators. The dataset was distributed among clients to simulate an FL environment with a non-IID data distribution. We then tested the FedAvg Algorithm with different numbers of clients (5, 10, 20, 50, 75) to determine the optimal setup for the Home Credit Risk Dataset.

B. NON-IDENTICAL DATA DISTRIBUTION ACROSS CLIENTS: SIMULATION

Once the clients are initialized, the data is split amongst the clients in a non-IID fashion. The home credit risk data set consisted of 3 lakh records. The level of data skewness across clients in the experiments is controlled by a hyperparameter α. A lower α creates a more significant smaller chunk of the dataset that is variedly distributed, while a higher α results in a more balanced distribution. non-IID data is also simulated by randomly varying the data each client receives. To assess the framework's performance under different levels of non-IID data, we conducted tests using the hyperparameter α as defined in Equation 11. Here, α_i is any number between 0 to 1, i is the client number and n is the number of clients.

$$\sum_{i=1}^{n} \alpha_i = 1 \tag{11}$$

$$\text{sizeofpartition}_i = \alpha_i \cdot \text{datasize} \tag{12}$$

In the simulation, there is no attribute skew, but only a skew in the probability distribution of the class labels and the data size per client mentioned in Equation 12. Since the non-IID ness of the data poses a considerable challenge in the Federated environment, the involvement of WGAN and Differential privacy comes to aid. WGANs use the Wasserstein distance, a reliable measure of dissimilarity between real and fake samples compared to traditional GANs, leading to more stable training.

C. UNIFIED DATA REPRESENTATION

Once the FL environment in simulated, we approach to preprocess the data. To leverage the strengths of GANs, which excel with continuous data, we aim to transform diverse client data—both numerical and categorical—into a continuous format. This involves using Min-Max scaling for numerical attributes and autoencoder for categorical attributes. The Home Credit Default Risk dataset contains 122 attributes, including 106 numerical and 16 categorical attributes.

1) SERVER-SIDE PROCESS

Each client sends the 122 attributes to the server in a key-value format, structured as (*categorical attribute name, list of unique values for that attribute*) and (*numerical attribute name, min and max values of that attribute*). These key-value pairs reveal the unique values present in each attribute per client while preserving individual privacy. The values associated with each key are transmitted to the server in randomized order, preventing attribution to specific clients. The server then aggregates the attribute-wise key-value pairs from all clients. Subsequently, it separates the numerical attributes from the categorical attributes, forwarding the numerical attributes to the Min-Max Module and the categorical attributes to the Autoencoder model.

1) ***Min-Max Module:*** The Min-Max Module determines the minimum and maximum values for each numerical attribute.
2) ***Autoencoder Model:*** After one-hot encoding of the categorical attributes, the autoencoder converts these binary representations into continuous values.

Finally, the server returns each numerical attribute's (Min, Max) values and a copy of the global autoencoder model to each client for categorical encoding.

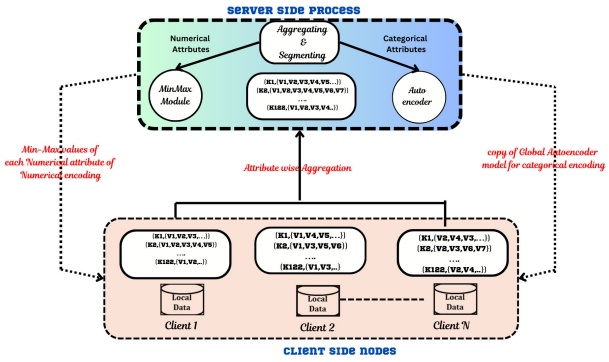


FIGURE 5. Towards effective GAN training: A unified approach for data representation using min-max module and autoencoder for converting mixed data to continuous data.

2) CLIENT-SIDE PROCESS:

a: NUMERICAL ENCODING

Each client utilizes the (Min, Max) pairs of numerical attributes to perform Min-Max Scaling on its data. Min-Max scaling is an essential preprocessing technique that enhances model performance by standardizing numerical features to a consistent scale. This technique adjusts the features to a standard range, typically between [0, 1]. The formula for Min-Max scaling is shown in Equation 13

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{13}$$

where:

- x is the original value,
- $x_{\min}$ is the minimum value in the feature,
- $x_{\max}$ is the maximum value in the feature.

b: CATEGORICAL ENCODING

Each client encodes their categorical attributes using a copy of the global autoencoder model. The entire process of unified data representation is shown in Figure 5.

Algorithm 1 outlines the process of the proposed standard approach for converting different clients' mixed type data into continuous data, and it is pictorially visualized in Figure 5.

D. FEDERATED WGAN WITH GAUSSIAN NOISE ADDED ADAGRAD - TO GENERATE PRIVACY PRESERVED SYNTHETIC DATA

With the global data representation phase complete, the data is now input into the Federated Wasserstein Generative Adversarial Network with an Adagrad optimizer enhanced by Gaussian noise (Fed-WGAN-Adagrad). During the training process, Gaussian noise is introduced at the optimizer level to mitigate the risks associated with membership inference and model inversion attacks. The proposed model is evaluated by varying privacy budgets, and a comprehensive test is conducted to assess the impact of membership inference on its performance.

Algorithm 1 Fed-DPSDG-WGAN: Unified Data Representation for Converting Clients Mixed Data Into Continuous Data Towards Effective GAN Training

Require: Mixed Datasets $D_1, D_2, \ldots, D_n$ from clients
Ensure: Continuous and scaled dataset $D_{\text{continuous}}$.

for each client $i = 1, 2, \ldots, n$ **do**
  Send all attributes to a server in key-value format:
  (attribute name, list of unique‖Min − Max values)
end for
Aggregate attribute-wise key-value pairs from all clients
Separate numerical attributes A_{num} and categorical attributes A_{cat}
for each numerical attribute $a_j \in A_{\text{num}}$ **do**
  Determine minimum and maximum values: $(\min_j, \max_j)$
end for
One-hot encode categorical attributes A_{cat}
Train Autoencoder AE on one-hot encoded data
Convert binary representations to continuous values:
$Z = \text{AE}(A_{\text{cat}})$
for each client $i = 1, 2, \ldots, n$ **do**
  Send back $(\min_j, \max_j)$ for each numerical attribute
  Send a copy of the global Autoencoder model for categorical encoding
end for
for each client $i = 1, 2, \ldots, n$ **do**
  for each numerical attribute a_j **do**
    Apply Min-Max Scaling using received $(\min_j, \max_j)$
  end for
  Use a copy of the global Autoencoder model to encode client-level categorical attributes into continuous values
end for
Return continuous datasets: $D_{\text{continuous}} = D_1 + D_2 + \ldots + D_n$

1) WGAN WITH GAUSSIAN NOISE ADDED ADAGRAD

The Wasserstein Generative Adversarial Network (WGAN) proposed by [52] offers several advantages over other GAN variants. First, the Wasserstein distance used as the loss function provides smoother, more meaningful gradients, enhancing optimization effectiveness, leading to better convergence and generating higher-quality samples. Second, WGAN allows for complete training of the critic, providing a reliable loss function for the generator. This stable setup simplifies the training process, making it similar to training other neural networks and eliminating the need to balance the generator and discriminator capacities carefully.

The WGAN's promising properties can suffice to a limit, but the addition of a Differentially Private Optimizer layer can prevent the model inversion attack.

Extensive research was conducted on a combination of differentially private optimizer layers in the WGAN

TABLE 4. WGAN with five gaussian noise addded optimizers that generates synthetic data close to real data.

GAN Variants	SGD	Adagrad	Adadelta	Adam	RMSProp
WGAN	×	✓	✓	×	✓

architecture to find the best trade-off between privacy and utility.

The generated synthetic data was evaluated using PCA plots to quantity similarity of the generated vs real data. Out of the tested five optimizers (SGD, Adagrad, Adadelta, Adam, RMSProp), each with added Gaussian noise. Adagrad, Adadelta, and RMSProp were identified as the top three based on data fidelity as shown in table 4. These three were then evaluated for privacy (Nearest Neighbor Adversary Accuracy Metric) and utility (via ML model performance). Adagrad with Gaussian noise was ultimately chosen for its optimal balance of fidelity, privacy, and utility.

4 All three optimizers achieved privacy scores above 0.5, indicating that they all successfully preserved privacy. However, when evaluating the utility of the generated synthetic data using machine learning models, the WGAN with Gaussian noise-added Adagrad optimizer achieved the highest accuracy of 82%. This outperformed both Adadelta and RMSprop, which achieved accuracies of 77.3% and 778.8%, respectively. Therefore, considering both privacy preservation and utility, the WGAN with Gaussian noise-added Adagrad optimizer is the best choice for generating differentially private synthetic data.

WGAN, with a mere Adagrad optimizer, is prone to inversion attacks, so in the proposed model, we added Gaussian noise to the Adagrad optimizer. When combined with Adagrad's adaptive learning nature, WGAN's strengths help faster convergence, yet the generated synthetic data is privacy-preserved. WGAN's discriminator and generator loss functions are defined in Equations 14 & 15

$$\text{DiscriminatorLoss}: L_D = -E_{x\sim D_{\text{real}}}[D(x)] + E_{z\sim N(0,I)}[D(G(z))] \quad (14)$$

$$\text{GeneratorLoss}: L_G = -E_{z\sim N(0,I)}(G(z)) \quad (15)$$

The Adagrad optimizer algorithm, along with Gaussian noise added to it is formulated as below Equation 16

$$\theta_{(t+i),i} = \theta(t,i) - \frac{\alpha}{G_{t,ii}+\epsilon} * ((g_{t,i}) + \eta_{t,i}) \quad (16)$$

where $\eta_{t,i}$ $N(0,\sigma^2)$ is the Gaussian noise sampled from a normal distribution with mean 0 and variance σ^2. The standard deviation is σ.

In order to establish a framework that adheres to differential privacy principles while preserving acceptable utility, we undertook an empirical investigation into the effects of different sensitivity values. The comprehensive analysis of these experiments is illustrated in the Results and Discussion section V-A.

The generator and discriminator of the Fed-DPSDG-WGAN is defined as follows:

1) ***Generator***: To generate synthetic data, the generator first uses a Dense layer to project the input, followed by a Reshape layer to prepare it for convolutional processing. BatchNormalization (momentum 0.8) standardizes the inputs for each mini-batch. An UpSampling layer then increases the dimensionality before a block of 1D Convolutional layers (kernel size 128, ReLU activation—Equation 17 [53]) further refines the data, with BatchNormalization (momentum 0.8) applied twice for stability.

$$ReLU(X) = max(0, X) \quad (17)$$

After processing through three Dense layers and a reshaped layer to create a 1×122 record (representing a single synthetic record with 122 values), the generator uses a final convolutional layer with a kernel size of 1 and a tanh activation function as shown in Equation 18 [53].

$$Tanh(x) = \frac{\exp(x) - \exp(-x)}{\exp(x) + \exp(-x)} \quad (18)$$

Because creating convincing synthetic records that resemble the original data is more complex than distinguishing real from fake, the generator is more sophisticated than the discriminator. The generator employs convolutional layers to capture the nuances of features in the tabular data, producing statistically accurate synthetic data. The model employs sparse cross-entropy loss and trains with a learning rate 10e-5.

2) ***Discriminator:*** The discriminator, responsible for distinguishing real from synthetic data, is trained on batches of 100. Consequently, both the real and synthetic subsets of the dataset have a size of 100×122, which serves as the discriminator's input shape. This input is then fed into a 1D Convolutional Network with Leaky ReLU activation (Equation 19 [53]).

$$LeakyReLU(X) = max(0.001 * X, X) \quad (19)$$

To prevent the "dying ReLU" problem, the Leaky ReLU activation function is used. Within a block repeated three times, BatchNormalization normalizes weights, and Dropout (0.25 probability) implements skip connections. The network concludes with a Flatten layer containing one neuron. This network is trained with a learning rate of 10e-5 to predict both the class of the input record using sparse cross-entropy loss and the sample's validity using binary cross-entropy loss.

The complete operational flow of the Federated WGAN, which incorporates an Adagrad optimizer with Gaussian noise injection, is described in Algorithm 2.

Once the clients and the Global Model (G) are initialized, the complete training process operates as follows:

1) *Model Initialization:* : A central server starts by initializing a base WGAN model with the noisy

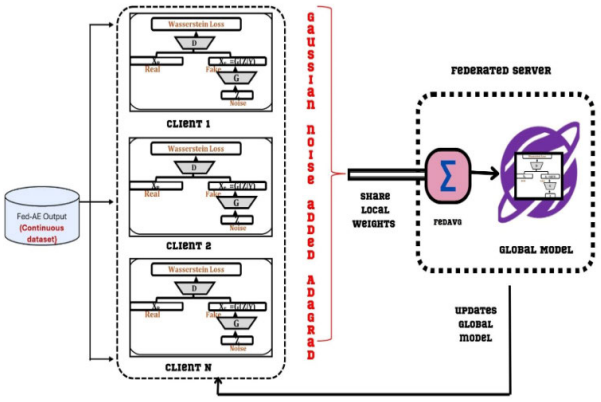


FIGURE 6. The proposed Federated WGAN with Gaussian noise added Adagrad optimizer architecture for generating differentially private synthetic data with better privacy-utility trade-off.

Adagrad optimizer, which it distributes to various client devices.

2) *Local Training:* Each client device uses its dataset to train the WGAN model locally. This training process involves nine epochs, enabling the model to learn from the unique data available on each device.
3) *Gradient Updates:* Rather than transmitting the entire dataset to the central server, each client sends only the updated gradients. This approach significantly enhances privacy, as raw data remains on the device.
4) *Aggregation:* The central server gathers these updates from all participating devices and averages the gradients to create a new version of the global model. This aggregation ensures the model benefits from diverse data sources while preserving data privacy.
5) *Iteration:* The system returns the updated global model to the client devices for further training. This cycle continues for several iterations until the model performs satisfactorily across various datasets.

Figure 6 depicts the architecture of the Federated WGAN with Gaussian noise added Adagrad optimization.

Since we iterated 100 cycles, this process generated 100*9 number of the client's fake records since the batch size is 100 and the number of iterations was nine. Therefore, it generated 4500 synthetic records for five clients, 9000 for ten clients, 22,500 for 25 clients, 45000 for 50 clients, 67500 for 75 clients, and 90,000 for 100 clients. Therefore, as the amount of training data increased, the number of synthetic records also increased

E. CTGAN - OVERSAMPLING THE SYNTHETIC DATA TO MAKE IT A BALANCED DATASET

CTGAN (Conditional Tabular Generative Adversarial Network) addresses imbalanced datasets by generating synthetic data using a conditional GAN architecture. Critically, CTGAN is "conditional," meaning it generates data based on specific features or classes, allowing it to focus on replicating the minority class and effectively augmenting it. Furthermore, CTGAN preserves the statistical properties and relationships within the original dataset in the synthetic data, avoiding distortions caused by more straightforward oversampling methods. This dual approach of enhancing minority class representation and maintaining statistical integrity provides more training examples while preventing overfitting and improving model generalization.

Oversampling improves model performance in several key ways including Addressing class imbalance, Enhancing model training, Improving metrics, and Representation of all classes.

In the proposed framework, after generating differentially private synthetic data using the federated WGAN, a copy of the global model is distributed to each client. Each client then trains its model using its local data to produce privacy-preserving synthetic data, which is often imbalanced. This imbalanced data at each client is subsequently balanced locally to improve model performance.

Algorithm 3 provides a detailed description of the process used to oversample privacy-preserved synthetic data with CTGAN.

Algorithm 2 Fed-DPSDG-WGAN: Differentially Private Synthetic Data Generation for Loan Defaulter Prediction via Federated Wasserstein GAN

Require: Pre-processed Datasets $D_1, D_2, \ldots, D_n$, Noise distribution $\mathcal{N}(0, \sigma^2)$, Learning rate α, Batch size m, Number of critic iterations n_{critic}

Ensure: Privacy Preserved Synthetic Datasets $S_1, S_2, \ldots, S_n$

Federated WGAN Training Process

Initialize global generator G, critic D, Adagrad optimizers
for global iteration $t = 1, 2, \ldots, T$ **do**
  for each client $i = 1, \ldots, n$ in parallel **do**
    Sample local batch x_i from transformed dataset D_i
    Sample noise $z_i \sim \mathcal{N}(0, I)$
    Critic loss: $L_D = D(G(z_i)) - D(x_i)$
    Generator loss: $L_G = -D(G(z_i))$
    Send local updates of G_i, D_i to server.
  end for
  Aggregate local model updates using FedAvg:
  $G = \frac{1}{n}\sum_{i=1}^{n} G_i,\ D = \frac{1}{n}\sum_{i=1}^{n} D_i$
  Fine Tune Global model parameters
end for
for each client $i = 1, \ldots, n$ **do**
    Generate synthetic data: $S_i = G(z_i)$
end for
  return Privacy Preserved Synthetic Data $S'_1, S'_2, \ldots, S'_n$

V. RESULTS AND DISCUSSION

The paper evaluates the effectiveness of the proposed framework by examining several key aspects:

Algorithm 3 Fed-DPSDG-WGAN: Oversampling Through CTGAN

Require: Synthetic Dataset $S'_1, S'_2, \ldots, S'_n$ from the clients

Ensure: Class balanced Privacy Preserved Synthetic Datasets $b_1, b_2, \ldots, _n$

// Oversampling the Synthetic data to get enhance Model performance

for each client $i = 1, 2, \ldots, n$ in parallel **do**

  Identify minority class in S'_i

  Train local CTGAN model G_i on S'_i

  Generate synthetic samples for minority class:

  $Sy'_i = G_i(z)$, where $z \sim \mathcal{N}(0, I)$

  Combine original and synthetic samples:

  $S'_i = S'_i + Sy'_i$

end for

Test the utility of the Fed-DPSDG-WGAN on the class-balanced synthetic dataset per client

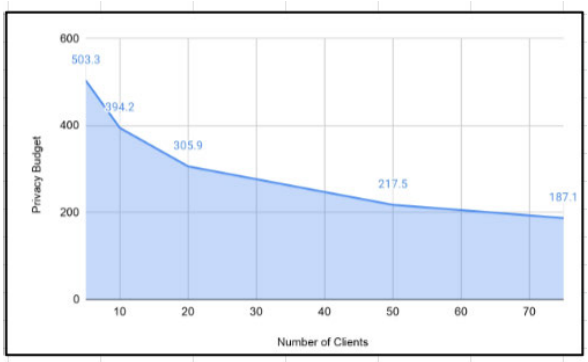


FIGURE 7. Trend of privacy budget reduction as client participation increases.

A. THE ROLE OF DATA VOLUME IN PRIVACY BUDGETING

Epsilon (ϵ) is a crucial parameter in differential privacy (DP) that determines the level of privacy protection provided. It regulates the amount of noise added to the data to safeguard individual information. A smaller ϵ indicates a stronger privacy guarantee, as it results in more noise being introduced, which can, however, compromise the accuracy of the results. In contrast, a larger ϵ offers a weaker privacy guarantee due to less noise being added, potentially enhancing accuracy but increasing the risk of exposing private data.The selection of ϵ is a critical decision that balances data utility with privacy concerns. Different levels of model privacy can be achieved by varying the sensitivity in the Gaussian noise-added optimizer. The privacy budget is a parameter that measures the balance between privacy protection and data utility. δ (delta) represents the probability that the privacy guarantee is violated. It is typically a small value.

Our observations indicated that with a δ value of 10e–5 and a noise multiplier of 0.3, the Fed-DPSDG-WGAN model - trained in 100 epochs with a batch size of 100 - showed a decreasing privacy budget as the number of clients increased. As the volume of training data increased, the reduction in the privacy budget underscored the need for greater collaboration among financial institutions to develop a robust model. Larger data sets demonstrated a better trade-off between privacy and utility.

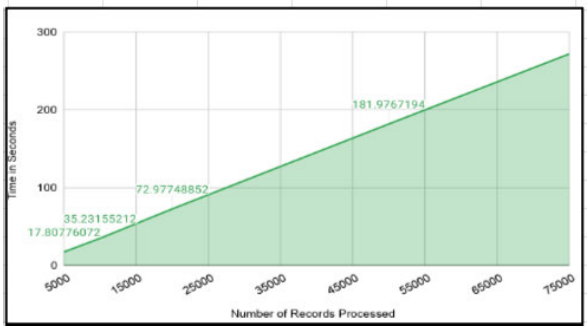


FIGURE 8. The trend of time taken with the number of records processed.

The sensitivity of the model's output directly influences the degree of differential privacy achieved. Consequently, since the epsilon parameter dictates the magnitude of Gaussian noise injected into the optimizer layer, precise calibration is needed to tailor it to the specific dataset characteristics.

In this study, we conducted a comprehensive empirical analysis across a range of epsilon values. Through rigorous evaluation, we determined that an epsilon value of 0.5 yielded an optimal balance between privacy preservation and data utility. As illustrated in Figure 7, while the privacy budget decreased, the accuracy of the machine learning model trained on synthetic data improved with increasing clients. A proper balance was achieved as the number of clients increased. Furthermore, our findings suggest that smaller datasets require a larger noise multiplier to maintain privacy. The observed increase in the privacy budget with more training data can be attributed to the enhanced generalization capabilities of the WGAN to generate differentially private synthetic data, making it more resilient against potential attacks.

B. EXECUTION TIME

The execution time for processing one epoch requires 17 seconds when using 25 clients and 5000 records, which escalates to 190 seconds with 75 clients. As anticipated, the duration of training increases with the data volume; this correlation is depicted in Figure 8. Efficient execution time in federated learning improves the utilization of distributed computational resources across clients, optimizing performance, and reducing operational costs. Crucially, maintaining this efficiency with an increasing number of clients demonstrates the scalability of the proposed framework. This scalability ensures the framework can effectively adapt to client participation and data volume growth without significant performance degradation.

C. TRAINING STABILITY

As we could see a trend of the utility of the synthetic data set deteriorating with an increasing privacy budget, we wanted to measure the part of increased clients and communication

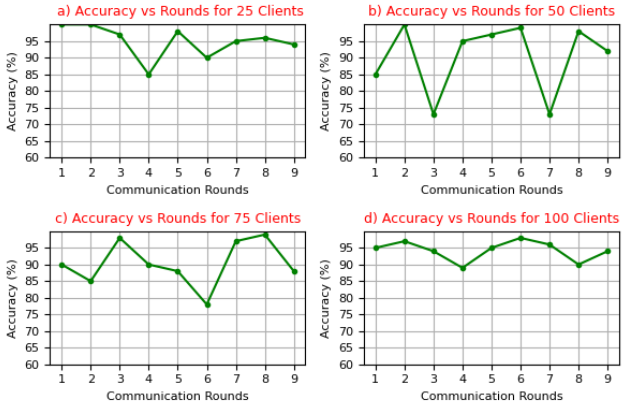


FIGURE 9. Accuracy vs. communication rounds trends across different client scalesa) 25 Clients, b) 50 Clients, c) 75 Clients, and d) 100Clients.

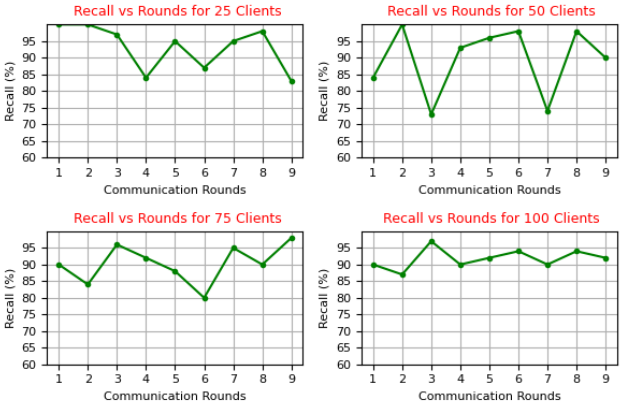


FIGURE 11. Recall vs. communication rounds trends across different client scales a) 25 Clients, b) 50 Clients, c) 75 Clients, and d) 100 Clients.

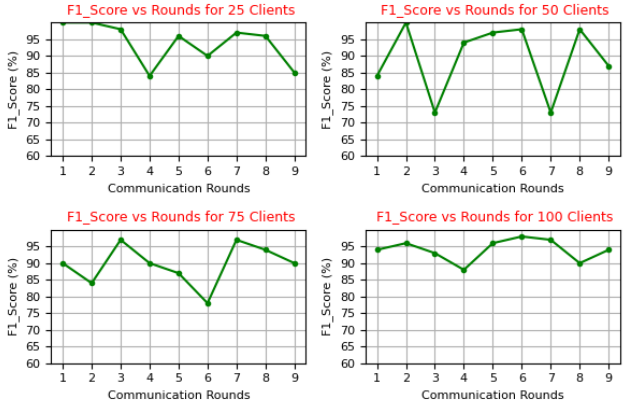


FIGURE 10. F1-Score vs. communication rounds trends across different client scales a) 25 Clients, b) 50 Clients, c) 75 Clients, and d) 100 Clients.

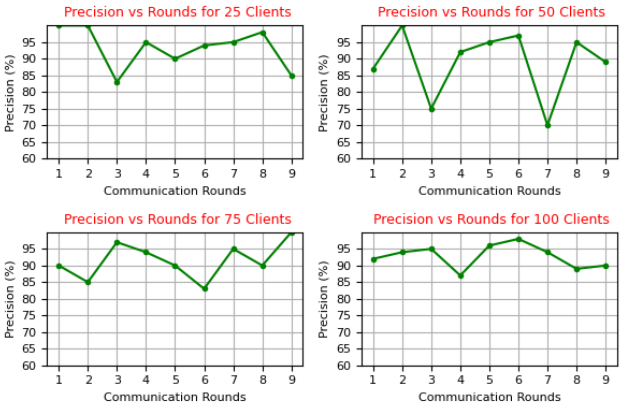


FIGURE 12. Precision vs. communication rounds trends across different client scales a) 25 Clients, b) 50 Clients, c) 75 Clients, and d) 100 Clients.

rounds in training stability. The Global Model (G) iterates through the clients for nine communication rounds. The training stability of the proposed Fed-DPSDG-WGAN framework after each communication round showcases trends for varying numbers of clients.

The initial pass of data led to overfitting in the models, primarily because the Global Model was trained on only 1,000 records per client and limited to ten local training epochs. Despite this, as the number of passes increased, key machine learning metrics - accuracy, F1 score, recall, and precision - began to stabilize. By the ninth pass, their performance improved significantly, reaching a commendable range of 92% to 95% as depicted in Figures 9 to 12. As more clients participate, the model encounters a wider variety of data, which helps it generalize better and reduces the risk of overfitting to specific data patterns from a smaller group of clients.

D. CENTRALIZED VS DECENTRALIZED APPROACH

It is clearly proven that increased data helps maintain utility in the synthetic dataset generated while maintaining privacy. Classification models utilizing machine learning are prevalent in credit risk assessment to differentiate loan defaulters from non-defaulters. Authors in the papers [54] and [55] evaluated various classifiers, including Logistic Regression (LR), Decision Trees (DT), and Random Forests (RF), in the context of Traditional credit risk assessment without anonymization. Similarly, when the loan defaulter classifiers were applied on the synthetic data, Random Forest performed well even on the synthetic data. Figure 13 clearly demonstrates that the performance of Random Forest outperformed the other classifiers, with an accuracy of 82% on the synthetic data generated using the Fed-DPSDG-WGAN. However, Privacy-preserving classification poses the challenge of balancing accuracy with safeguarding data. To address this, the Fed-DPSDG-WGAN framework was developed and evaluated in both centralized and distributed settings, showcasing its effectiveness in achieving this balance.

The synthetic data generated using the framework is tested using multiple ML models. The ML models are trained on the actual dataset and tested on the synthetic data. If the model performs well, it means that the synthetic data replicates the original data well. Still, since the synthetic dataset is generated in a Federated and DP environment, it is privacy preserved. Research has shown that Federated Random Forests (FRF) excel in handling challenges like limited local data, dataset heterogeneity, and imbalanced classes while achieving accuracy comparable to centralized models. Random Forest's inherent ability to identify and prioritize key

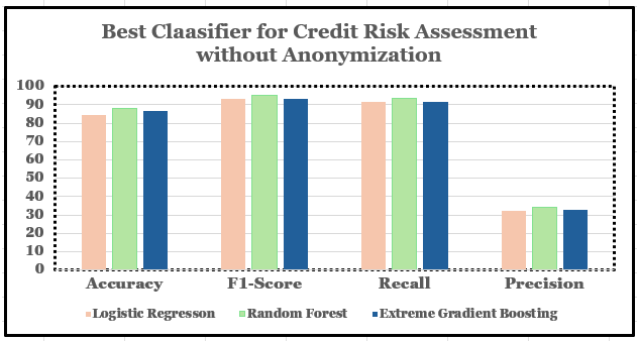


FIGURE 13. Performance comparison of classifiers for credit risk assessment in a non-privacy-preserving setting.

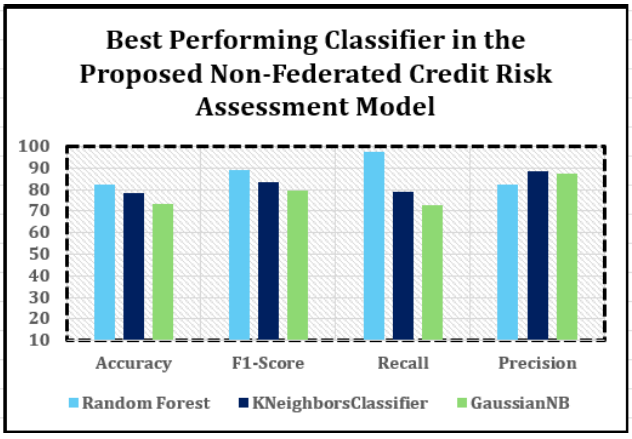


FIGURE 14. Best performing classifier in the proposed non-federated credit risk assessment model.

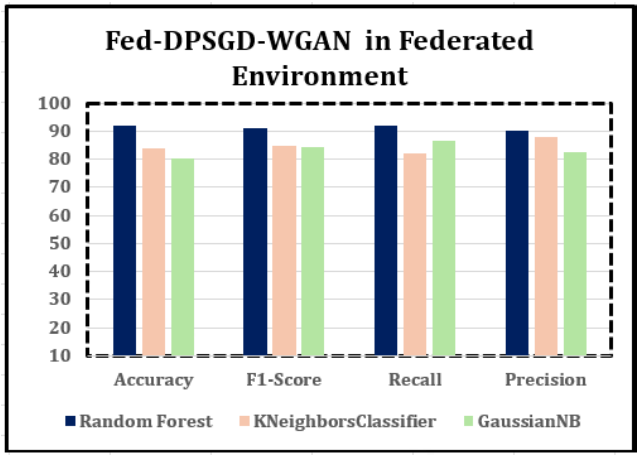


FIGURE 15. Best performing classifier of the proposed framework in federated environment.

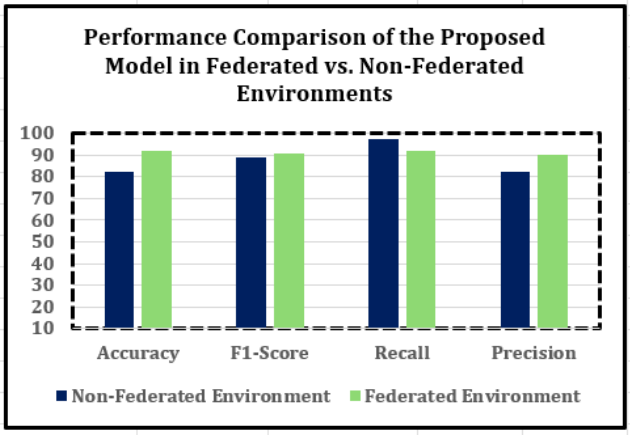


FIGURE 16. Performance comparison of the proposed model in federated vs non-federated environments.

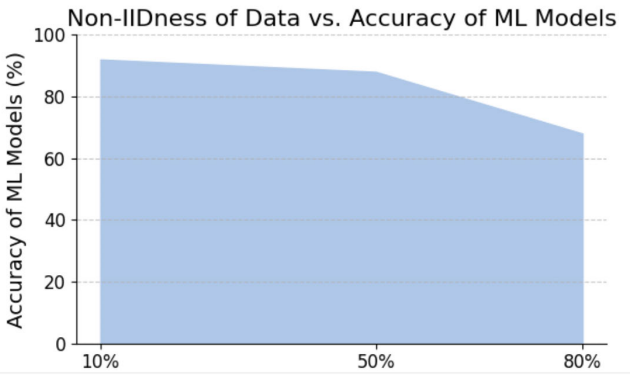


FIGURE 17. Impact on utility with varying degrees of non-IIDs for 10 clients.

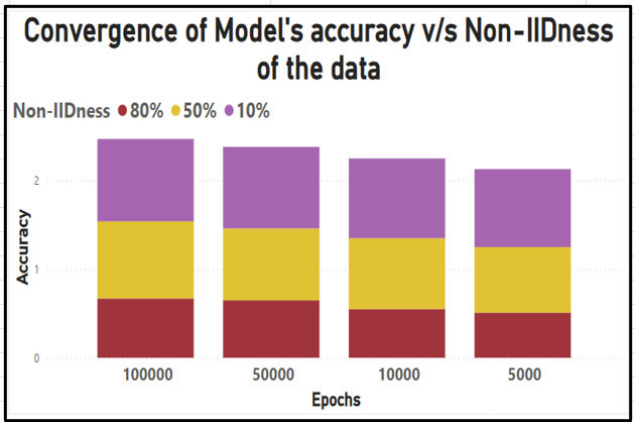


FIGURE 18. Impact of data non-IID-IIDness on model accuracy convergence.

features further enhances this robust performance, leading to improved interpretability and model refinement. [56], [57]. These strengths have positioned Random Forest, as illustrated in Figure 14 and 15, as the most effective classifier in our proposed framework, outperforming other classifiers such as KNeighbors and GaussianNB.

The proposed federated framework surpasses centralized approaches in both privacy and model performance. Privacy is strengthened by keeping data localized on user devices and sharing only model updates for aggregation. Model performance is enhanced because the model benefits from training on diverse data from multiple banks. Figure 16 illustrates that the performance metrics of the federated approach are consistently around 90% or higher when compared to the centralized approach.

TABLE 5. Comparative analysis of the proposed model with state-of-the-art approaches.

Algorithm	Privacy Preservation	Synthetic Data Generation	Handles Incomplete data	Method Used	Accuracy
Fed-TGAN [36]	✓	✓	×	Federated GAN with VGM	
HT-Fed-GAN [34]	✓	✓	×	Federated GAN with VB-GMM	
FLIGAN [31]	✓	✓	✓	Federated GAN w/ classwise sampling & node grouping	73
Fed-DPSDG-WGAN (Proposed)	✓	✓	✓	**Autoencoder & federated WGAN with Gaussian noise added Adagrad**	**95**

TABLE 6. Ablation study: Evaluating the impact of key components in the proposed model.

Architecture Component Description	Models	non-IID ness	Accuracy
Proposed Framework with three components	**Autoencoder + Fed WGAN + CTGAN**	Yes	**95**
By Removing Federated Learning	Autoencoder + WGAN + CTGAN	No	82.3
By Removing CTGAN	Autoencoder + WGAN	No	90.3

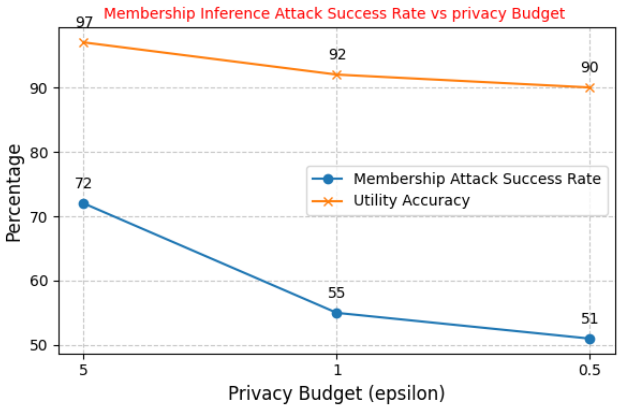


FIGURE 19. Privacy Budget Vs Membership inference attack success rate.

E. IMPACT OF VARYING DEGREES OF NON-IIDNESS ON THE SYNTHETIC DATASET GENERATION:

As the non-IID nature of the data increased, the model required more epochs to achieve consistent convergence with accuracy exceeding. This is because the performance of the Fed-Avg algorithm degrades in finding the optimal weights to create generalized synthetic data as client data becomes more dissimilar. In highly non-IID scenarios, WGANs face challenges in capturing the full diversity of the client data. Nevertheless, WGANs demonstrated superior modeling of non-IID data compared to other GAN variants. We could see that as the non-IID ness of the train data increases, the performance of ML models tested on the synthetic data generated drops. If the train data is 50% non-IID the synthetic data is still maintaining utility, as shown in figure 17.

From Figure 18, it is clear that as the non-IIDness of the data increases, the utility of the generated synthetic data decreases. However, the performance can be compensated, and the utility of the data can be improved by training with a huge quantity and variety of data, which can be observed in Figure 18.

F. MEMBERSHIP INFERENCE ATTACK SUCCESS RATE VS PRIVACY BUDGET

To evaluate the framework's robustness, we conducted a functional membership inference test. We merged the generated synthetic dataset with a subset of samples from the original training data, labeling each sample as either "real" or "fake." If a machine learning model trained on this combined dataset could accurately identify the fake samples with an accuracy exceeding 50%, the synthetic dataset was deemed not sufficiently privacy-preserved.

We simulated a membership inference attack on the synthetic dataset generated using 10 clients. When this data was used to train and test a Random Forest Classifier, we observed a positive correlation between the privacy budget and the membership attack success rate. As the privacy budget increased, the sucess rate of MIA decreased. Simultaneously, the accuracy of the model, used to measure utility, exhibited a downward trend as the privacy of the model increased. However, we achieved a desirable privacy-utility trade-off at $\epsilon = 0.5$, where considerable utility (90% accuracy) was maintained while ensuring a reasonable level of data privacy as demonstrated in Figure 19.

G. ABLATION STUDY

In addition to understanding the framework's performance and robustness, an ablation study was carried out to assess the effectiveness of each component, identify areas for improvement, and promote privacy-preserving machine learning in finance. Table 6 presents a concise ablation study, illustrating the effects of different components within the proposed framework.

The proposed model, which incorporates federated components and is trained on non-IID data, achieves an impressive accuracy of 95%. This demonstrates that combining the autoencoder, federated WGAN, and CTGAN effectively manages non-IID data while delivering accurate predictions.

Removing federated learning reduces accuracy to 82.3% because it limits data access and compromises privacy. Without federated learning, the model struggles to generalize effectively from diverse data, which hinders its performance.

Removing CTGAN leads to a 90.3% accuracy. While this is still relatively high, it's crucial to recognize that accuracy alone can be misleading with imbalanced data. The model may mainly excel at predicting the majority class, obscuring the poor performance of the minority class. Therefore, this accuracy figure doesn't guarantee overall good performance. In summary, the ablation study highlights the critical importance of federated learning and specific generative modeling techniques (WGAN and CTGAN) in achieving high accuracy within a distributed environment characterized by non-IID data. It provides compelling evidence for the

effectiveness of the proposed framework and justifies its design choices.

H. STATE OF ART METHODOLOGIES COMPARISON

Table 6 showcases state-of-the-art methodologies for privacy-preserving synthetic data generation, including Fed-TGAN, HT-Fed-GAN, and FLIGAN, each innovating with GAN-based techniques to balance privacy-utility trade-off.

In contrast to the above methodologies, the proposed framework demonstrates superior performance with an accuracy of 95%. This improvement highlights its effectiveness in generating high-quality synthetic data while addressing the challenges of privacy preservation and incomplete data handling more robustly than existing solutions.

Overall, while Fed-TGAN and HT-Fed-GAN are effective for privacy-preserving synthetic data generation, they fall short in managing incomplete data. FLIGAN attempts to tackle this issue but at the cost of lower accuracy. The proposed work outperforms these state-of-the-art methodologies significantly, providing a more effective solution for generating synthetic data in a privacy-preserving manner.

VI. CONCLUSION

'This paper presented a novel and practical federated learning framework for privacy-preserving loan defaulter prediction. The framework effectively tackles several key challenges by incorporating an Autoencoder, a noise-enhanced Federated Wasserstein Generative Adversarial Network, and an oversampling technique. These include preserving privacy, maintaining model performance, ensuring training stability, handling non-independent and Identically Distributed data, mitigating susceptibility to membership inference attacks, model inversion attacks, and finally addressing class imbalance issues. Specifically, the federated approach facilitates collaboration across multiple banks, mitigating data heterogeneity and promoting robust model training by exchanging only model parameters, not raw data, thereby enhancing privacy and enhanced predictive accuracy in credit risk evaluation. Adding Gaussian noise to the Adagrad optimizer further strengthens privacy by mitigating model inversion attacks. Comprehensive experimental results demonstrate the framework's superior performance in the federated setting, achieving 95% accuracy compared to 82% in the non-federated environment.

To effectively safeguard client data confidentiality, the proposed Fed-DPSDG-WGAN architecture incorporates a robust differential privacy optimizer layer, ensuring data localization and mitigating security risks associated with centralized processing. Data synchronization is achieved through a consensus-based attribute generation mechanism across participating clients. Our empirical evaluation, conducted with 300,000 records across 100 clients, demonstrated a 3.5-minute latency. Projecting to a 10-15 fold data increase, we anticipate a maximum latency of one hour, which is deemed acceptable for periodic execution. To further enhance the practical deployment of the Fed-DPSDG-WGAN framework, future research will prioritize resource efficiency improvements, specifically through bandwidth optimization strategies like selective model updates and adaptive aggregation frequencies, ensuring the framework's efficacy and scalability for large-scale privacy-preserving synthetic data generation

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